



Graphic Organizer



Team 7: Brian Ngo, Omid Ghofrani, Prince Mundy

Topic: Vaccine Technologies

Paper 1:

[Nanoparticle-based DNA vaccine protects against SARS-CoV-2 variants in female preclinical models](#)

(Guimaraes, L.C., Costa, P.A.C., Scalzo Júnior, S.R.A. et al. Nanoparticle-based DNA vaccine protects against SARS-CoV-2 variants in female preclinical models. Nat Commun 15, 590 (2024). <https://doi.org/10.1038/s41467-024-44830-1>)

Similarities:

- **Use of Lipid Nanoparticles (LNPs):**
Both studies employ LNPs for vaccine delivery to improve stability and immune response
- **Strong Immune Activation:**
Both vaccines induce robust immune responses, with significant T-cell activation and neutralizing antibody production
- **Preclinical Animal Testing:**
Both studies evaluate vaccine efficacy in mouse models and non-human primates

Paper 2:

[An mRNA-based T-cell-inducing antigen strengthens COVID-19 vaccine against SARS-CoV-2 variants](#)

(Tai, W., Feng, S., Chai, B. et al. An mRNA-based T-cell-inducing antigen strengthens COVID-19 vaccine against SARS-CoV-2 variants. Nat Commun 14, 2962 (2023). <https://doi.org/10.1038/s41467-023-38751-8>)

What types of experiments were done? • In Vitro Transfection Assays • In Vivo Biodistribution Studies • Animal Vaccine Studies • Immune Response Analysis • Histopathological and Viral Load Analysis		What types of experiments were done? • mRNA Vaccine Expression and Processing • Immunogenicity in Mice • SARS-CoV-2 Challenge in Mice • Dual Immunization Study • Non-Human Primate Study
What were the results that were achieved? • Enhanced DNA Delivery: The LNP B4 formulation demonstrated superior transfection efficiency in antigen-presenting cells, fibroblasts, and endothelial cells • Effective Protection Against SARS-CoV-2: Mice vaccinated with the LNP-based DNA vaccine (LNP-HPS) showed 85% survival after a lethal challenge with the Gamma (P.1) variant, compared to 100% lethality in control groups • Reduced Viral Load: Immunized mice had significantly lower SARS-CoV-2 viral loads in the lungs, comparable to those vaccinated with the Pfizer-BioNTech mRNA vaccine (BNT-mRNA) • Strong Immune Response • Improved Safety Profile	Differences: • Mechanism of Action Paper 1: Targets T-cell epitopes to create a long-lasting immune response Paper 2: Encodes the HexaPro spike protein to stimulate an immune response • Variant Protection Strategy Paper 1: Optimizes T-cell memory to combat SARS-CoV-2 mutations Paper 2: Directly encodes spike proteins from variants for broad neutralizing activity • Dual Immunization Approach Paper 1: Tested dual immunization (T-cell antigen + RBD antigen) Paper 2: Focused on single-component vaccination • Stability and Storage Paper 1: Requires ultra-cold storage but allows faster vaccine updates Paper 2: More stable at room temperature, advantageous for global distribution	What were the results that were achieved? • Induced strong T-cell activation, improving vaccine protection against Beta and Omicron variants • Dual immunization (T-cell antigen + RBD) significantly reduced viral loads in lungs and improved neutralization • Maintained protection despite SARS-CoV-2 mutations, showing enhanced long-term immunity